SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO1 ASSESSMENT TOOL 2 – Agile Sprint Planning & User Story Workshop

Course Code:	CSA10 / CSA1063	**Course Title:**	Software Engineering
CO Assessed:	CO1-AT2	**Assessment:**	Assessment Tool 2 – **Agile Sprint Planning & User Story Workshop**
Weightage:	20%	**Total Marks:**	25
CO1 Statement:	*Apply advanced methodologies and agile project management techniques to manage software projects effectively*		
Student Name:	MS NAVEEN	**Reg. No.:**	192371002
Date:	26/07/2026	**Duration:**	60 minutes

Code of Conduct

I NAVEEN MS(192371002 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: NAVEEN

Annexure A – Agile Sprint Planning & User Story Workshop

Instructions to Students:

Each student is assigned an individual software project problem statement. Based on the given problem statement, perform an **Agile Sprint Planning and User Story Workshop** by applying Agile software engineering practices. Prepare a comprehensive sprint planning document covering the following aspects:

Based on the given problem statement, perform an **Agile Sprint Planning and User Story Workshop** by applying Agile software engineering practices. Prepare a comprehensive sprint planning document covering the following aspects:

1. **Identify the Project Vision and Stakeholders** ○ Define the project's vision, objectives, and identify all key stakeholders involved.
2. **Create Epics and User Stories** ○ Break down the requirements into Epics and formulate well-structured User Stories using the format: *As a <user>, I want <feature>, so that <benefit>.*
3. **Define Acceptance Criteria** ○ Write clear and measurable acceptance criteria for each selected user story using the Given–When–Then format or equivalent.
4. **Prioritize the Product Backlog** ○ Organize and prioritize the product backlog based on business value, customer needs, and project dependencies.
5. **Plan the Sprint Backlog** ○ Select high-priority user stories for the first sprint and divide them into actionable development tasks.

6. **Estimate Effort Using Story Points** ○ Assign story points to each selected user story using an appropriate estimation technique (e.g., Fibonacci sequence or Planning Poker) and justify your estimates.
7. **Present the Sprint Goal and Expected Deliverables** ○ Define a clear sprint goal and describe the expected sprint deliverables, including the features to be completed at the end of the sprint.

Rubrics for Calculation

Criteria	Excellent (5 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Project Vision & Stakeholder Identification	Project vision is clear, comprehensive, and aligned with the problem statement. All relevant stakeholders and their roles are correctly identified.	Vision is clear with minor omissions. Most stakeholders are correctly identified.	Vision is partially defined. Some stakeholders are missing or incorrectly identified.	Vision is unclear or incomplete. Stakeholders are poorly identified or missing.
Epics & User Stories	Epics are well-defined and user stories are complete, follow Agile format, and fully capture functional requirements.	Epics and user stories are mostly complete with minor formatting or requirement issues.	User stories are partially complete with limited Agile structure.	User stories are incomplete, unclear, or do not follow Agile practices.
Acceptance Criteria	Acceptance criteria are specific, measurable, testable, and aligned with all user stories.	Acceptance criteria are mostly clear and testable with minor gaps.	Acceptance criteria are partially defined and lack clarity or completeness.	Acceptance criteria are missing or not aligned with user stories.
Product Backlog Prioritization & Sprint Planning	Product backlog is logically prioritized, sprint backlog is realistic, and task breakdown is complete.	Backlog prioritization and sprint planning are mostly appropriate with minor issues.	Prioritization or sprint planning is partially complete with limited justification.	Backlog prioritization and sprint planning are poorly organized or incomplete.
Story Point Estimation, Sprint Goal & Presentation	Story point estimation is well justified, sprint goal is clear, deliverables are realistic, and presentation is professional and well organized.	Estimation and sprint goal are mostly appropriate with minor inconsistencies. Presentation is clear.	Estimation or sprint goal lacks justification. Presentation is adequate but incomplete.	Estimation is unrealistic or missing. Sprint goal, deliverables, and presentation are unclear or incomplete.

Criteria	Mark
Project Vision & Stakeholder Identification	/5
Epics & User Stories	/5
Acceptance Criteria	/5
Product Backlog Prioritization & Sprint Planning	/5

Story Point Estimation, Sprint Goal & Presentation	/5
TOTAL	/25

Annexure A – Agile Sprint Planning & User Story Workshop

Project Title
EV Charging Station Management System

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1. Identify the Project Vision and Stakeholders

Project Vision
The EV Charging Station Management System is a web and mobile-based application that helps electric vehicle (EV) owners easily locate charging stations, book charging slots, make online payments, and monitor charging status in real time. The system also allows station operators to manage charging points, monitor usage, and generate reports. The project aims to reduce waiting time, improve charging efficiency, and encourage the adoption of electric vehicles.

Project Objectives

- **Provide an easy way to locate nearby EV charging stations.**
- **Allow users to reserve charging slots in advance.**
- **Enable secure online payment.**
- **Display real-time charging status.**
- **Help station operators manage charging stations efficiently.**
- **Generate reports for monitoring station usage and revenue.**

Stakeholders

Stakeholder	Role
EV Vehicle Owner	Searches, books, and pays for charging services
Charging Station Operator	Manages charging stations and bookings

Stakeholder	Role
Administrator	Manages users, stations, and system settings
Payment Gateway Provider	Processes secure online payments
Maintenance Engineer	Maintains charging equipment
Project Manager	Plans and manages project activities
Scrum Master	Ensures Agile practices are followed
Development Team	Designs and develops the application
QA/Test Engineer	Tests application quality
Government/Electric Utility	Provides charging regulations and electricity support

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2. Create Epics and User Stories

Epic 1: User Account Management

User Story 1

As an EV owner, I want to register an account so that I can use the charging services.

User Story 2

As an EV owner, I want to log in securely so that my personal information is protected.

Epic 2: Charging Station Search

User Story 3

As an EV owner, I want to search nearby charging stations so that I can find the nearest available charger.

User Story 4

As an EV owner, I want to view charging station details so that I can know charger type, price, and availability.

Epic 3: Slot Booking
User Story 5
As an EV owner, I want to book a charging slot so that I can avoid waiting in queues.

User Story 6
As an EV owner, I want to cancel my booking so that I can change my travel plan if needed.

Epic 4: Payment System
User Story 7
As an EV owner, I want to pay online so that I can complete my booking securely.

Epic 5: Charging Management
User Story 8
As an EV owner, I want to monitor charging status so that I know when my vehicle is fully charged.

Epic 6: Administration
User Story 9
As an administrator, I want to manage charging stations so that the station information remains accurate.

User Story 10
As an administrator, I want to generate reports so that I can analyze charging station usage and revenue.

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3. Acceptance Criteria
User Story 1 – User Registration
Given
The user is on the registration page.

When
The user enters valid information and clicks Register.
Then

- **Account is created successfully.**
- **Success message is displayed.**
- **User can log in.**

User Story 2 – Login
Given
The user has a registered account.
When
The user enters valid username and password.
Then

- **Login is successful.**
- **Dashboard is displayed.**

User Story 3 – Search Charging Station
Given
The user is logged in.
When
The user enters a location.
Then

- **Nearby charging stations are displayed.**
- **Distance and availability are shown.**

User Story 4 – View Station Details
Given
Charging stations are displayed.
When
The user selects a station.
Then

- **Charger type**
- **Charging cost**

- Available slots
- Operating hours

are displayed.

User Story 5 – Book Charging Slot
Given
A charging slot is available.
When
The user selects the slot and confirms booking.
Then

- **Booking is confirmed.**
- **Booking ID is generated.**
- **Confirmation notification is sent.**

User Story 6 – Cancel Booking
Given
The booking exists.
When
The user clicks Cancel Booking.
Then

- **Booking is cancelled.**
- **Slot becomes available for others.**

User Story 7 – Online Payment
Given
Booking is confirmed.
When
The user completes payment.
Then

- **Payment is successful.**
- **Digital receipt is generated.**

User Story 8 – Charging Status

Given
Vehicle is connected.
When
Charging starts.
Then

- **Charging percentage is updated.**
- **Estimated completion time is displayed.**

User Story 9 – Manage Charging Stations
Given
Administrator logs in.
When
Administrator adds or edits station details.
Then

- **Updated station information is saved.**

User Story 10 – Generate Reports
Given
Charging data is available.
When
Administrator selects Generate Report.
Then

- **Revenue report**
- **Booking report**
- **Usage report**

are generated.

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4. Product Backlog Prioritization

Priority	User Story	Business Value
High	User Registration	Very High
High	User Login	Very High
High	Search Charging Station	Very High

Priority	User Story	Business Value
High	View Station Details	High
High	Book Charging Slot	Very High
High	Online Payment	Very High
Medium	Charging Status	High
Medium	Cancel Booking	Medium
Medium	Manage Charging Stations	Medium
Low	Generate Reports	Low

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5. Sprint Backlog

Sprint Duration

2 Weeks

Sprint 1 User Stories

- User Registration
- User Login
- Search Charging Stations
- View Station Details
- Book Charging Slot

Development Tasks

User Registration

- Design registration page
- Validate user inputs
- Store user information in database
- Password encryption

User Login

- Login interface
- Authentication module
- Session management

Search Charging Station

- GPS integration
- Fetch charging stations
- Display nearest stations

Station Details

- Display charger type
- Display charging cost
- Show slot availability

Book Slot

- Booking interface
- Slot availability check
- Booking confirmation

Sprint Backlog Table

Task	Assigned To
Registration Page	Frontend Developer
Login Module	Backend Developer
Database Design	Database Administrator
GPS Integration	Backend Developer
Station Search UI	Frontend Developer
Booking Module	Backend Developer
API Development	Backend Developer
Testing	QA Engineer

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6. Story Point Estimation

Estimation Technique

Fibonacci Sequence (1, 2, 3, 5, 8, 13)

User Story	Story Points	Justification
User Registration	3	Simple form with validation
User Login	3	Authentication logic
Search Charging Station	8	GPS integration and database search
View Station Details	5	Data retrieval and UI display
Book Charging Slot	8	Booking validation and slot management

Total Story Points
3 + 3 + 8 + 5 + 8 = 27 Story Points

Justification

- **3 Points: Low complexity with basic forms and validation.**
- **5 Points: Medium complexity involving data retrieval and display.**
- **8 Points: High complexity due to GPS integration, booking logic, and availability checks.**

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7. Sprint Goal and Expected Deliverables

Sprint Goal

Develop the core functionalities of the EV Charging Station Management System by enabling user registration, secure login, charging station search, station information display, and charging slot booking.

Expected Sprint Deliverables

- **✓ User Registration Module**
- **✓ User Login Module**
- **✓ Charging Station Search**
- **✓ Charging Station Details Page**
- **✓ Charging Slot Booking Module**

- ✓ Database Tables
- ✓ REST APIs
- ✓ Basic UI Screens
- ✓ Unit Testing and Bug Fixes

Sprint Summary

Sprint	Sprint 1
Duration	2 Weeks
Sprint Goal	Develop user management and charging station booking features
Story Points	27
Team Members	Product Owner, Scrum Master, Frontend Developer, Backend Developer, Database Administrator, QA Engineer
Deliverables	Registration, Login, Station Search, Station Details, Slot Booking

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Agile Workflow

Project Vision

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Product Backlog

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Sprint Planning

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Sprint Backlog

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Design & Development

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Testing

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Sprint Review

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Sprint Retrospective
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Working Product Increment
This version is tailored specifically to the EV Charging Station Management System and covers all seven required Agile Sprint Planning components in a clear, structured format suitable for academic submission.